Adaptive Multi-Objective Chaotic S-Box Design for Lightweight and Secure IoT Communication

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Abstract

Substitution boxes (S-boxes) are nonlinear components used in many symmetric ciphers, but their security properties must be evaluated independently of any complete encryption scheme. We study a deterministic Fisher–Yates permutation generator driven by a coupled two-dimensional Logistic map and compare NSGA-II with and without periodic Flower Pollination Algorithm (FPA) perturbations. The optimizer uses three independent objectives: largest Lyapunov exponent, minimum coordinate nonlinearity, and differential uniformity. Avalanche and linear-approximation metrics are calculated after optimization. In five paired simulation seeds, mean selected minimum nonlinearity was 102.0 (SD 0.00) for FPA and 102.0 (SD 1.41) for NSGA-II-only; mean differential uniformity was 10.0 (SD 0.00) for both variants. Exact paired tests with Holm correction detected no statistically supported differences in the reported metrics. The corrected Lyapunov estimates are on the expected scale (approximately 0.67–0.69 for the selected solutions), not the previously reported values near 16.8. The final selected LE range was 0.666–0.682. The study is simulation-only: no complete cipher or target-device timing is evaluated, and the results do not establish suitability for deployment.

Keywords: Multi-objective optimization; NSGA-II; Chaotic Map; Substitution Box; Lightweight Cryptography; Internet of Things.

1 Introduction

Substitution boxes (S-boxes) provide the nonlinear component in many symmetric-key ciphers. Their evaluation commonly includes Boolean nonlinearity, avalanche behavior, differential uniformity, and linear approximation bias; these criteria capture distinct structural properties and should not be conflated with the security of a complete cipher [1, 2] [3, 5]. Chaotic-map-driven permutation generation and evolutionary search have both been studied as construction approaches [6, 7] [9].

Prior chaotic S-box studies have explored map-driven permutation construction and evolutionary search, but comparisons depend on the specific map, objective definitions, and evaluation protocol. For this reason, we focus here on a controlled, reproducible comparison of two search variants under the same implementation and seed schedule rather than claiming superiority over the literature.

We implement an NSGA-II search on parameters of a modulo-one coupled Logistic map and compare it with the same search augmented by periodic Flower Pollination Algorithm (FPA) perturbations. The search uses three independent objectives: largest Lyapunov exponent, minimum coordinate nonlinearity, and differential uniformity. We evaluate SAC, BIC-SAC deviation, and linear approximation probability after optimization; they are not part of the fitness vector. Our five-seed comparison is exploratory and does not demonstrate a statistically supported advantage for FPA.

The rest of the paper is organised as follows. Section 2 develops the 2D Logistic map, the map-to-permutation construction, the multi-objective fitness formulation, and the NSGA-II–FPA hybrid. Section 3 reports the corrected Pareto search, post-hoc S-box metrics, and the paired five-seed ablation. Section 4 interprets the simulation-only evidence and its limitations. Section 5 summarises the findings and identifies the experiments required next.

2 Materials and Methods

Notation. We treat an 8-bit S-box as a permutation $S : \{0,1\}^8 \to \{0,1\}^8$. Let $x_n, y_n \in [0,1)$ be the state variables of the modulo-one 2D Logistic map. The optimizer searches phase genes $p_x, p_y \in [0,1)$ and maps them to valid initial states by $x_0 = \sin^2(\pi p_x)$ and $y_0 = \sin^2(\pi p_y)$. The control parameters satisfy $\mu \in [3.57, 4.0)$ and $\gamma \in [0, 0.5)$. Let χ_{LE} denote the largest Lyapunov exponent, NL the minimum per-coordinate nonlinearity, DU the differential uniformity, LAP the linear approximation probability, SAC the strict avalanche criterion, and $\mathrm{BIC}_{\mathrm{SAC}}$ the BIC-SAC deviation from 0.5. These metrics follow the canonical S-box evaluation framework of [1, 2] [3, 4] [5, 2].

Two-dimensional Logistic map. The 2D Logistic map is a coupled quadratic iteration with a single nonlinearity parameter μ and a coupling parameter γ:

$$\begin{aligned} x_{n+1} &= \mu \cdot x_n \cdot (1 - x_n) + \gamma \cdot y_n \\ y_{n+1} &= \mu \cdot y_n \cdot (1 - y_n) + \gamma \cdot x_n. \end{aligned} \tag{1}$$

Equation 1 defines the modulo-one map used in this simulation. The parameter μ controls the coordinate nonlinearity, and γ couples the coordinates. When $\gamma = 0$, the system reduces to two independent one-dimensional logistic maps; selected solutions are therefore checked for proximity to this boundary.

We estimate the largest Lyapunov exponent by iterating a unit tangent vector v_n under the Jacobian and renormalizing after each step:

$$v_{n+1} = \frac{J(x_n, y_n) v_n}{\|J(x_n, y_n) v_n\|}, \qquad \chi_{\mathrm{LE}} \approx \frac{1}{N} \sum_{n=0}^{N-1} \log \|J(x_n, y_n) v_n\|, \tag{2}$$

where J is the Jacobian of the map and an initial transient is excluded. For $\gamma = 0$ and $\mu = 4$, the expected largest exponent is $\log 2$; this provides a numerical sanity check on the implementation.

Map → permutation (Fisher–Yates driver). We construct an 8-bit permutation by applying Fisher–Yates to the identity permutation and using successive x-coordinates of the modulo-one map as swap keys. After burning a transient of $T_0 = 500$ map iterations, one map step

supplies each key $k_i \in [0, 1)$ for

$$\begin{aligned} \text{perm} &= [\,0, 1, \ldots, 255\,]; \\ &\text{for } i = 255 \text{ down to } 1: \\ &\quad j = \lfloor k_i \cdot (i+1) \rfloor,\ \text{perm}[i] \leftrightarrow \text{perm}[j], \end{aligned} \tag{3}$$

Equation 3 produces a deterministic permutation of length 256 with no collisions because Fisher–Yates preserves bijectivity by construction. The chaotic keys are highly sensitive to the parameters (x_0, y_0, μ, γ), so a small parameter perturbation yields a structurally different permutation.

Multi-objective fitness. We minimize the following three-objective vector:

$$\mathbf{f}(S; x_0, y_0, \mu, \gamma) = \big(-\chi_{\text{LE}},\ -\text{NL}_{\min}(S),\ \text{DU}(S)\big), \tag{4}$$

The three objectives are independent quantities in the implemented fitness. SAC, BIC-SAC deviation, and LAP are evaluated after search and do not affect selection. No runtime or energy measurement is included in the fitness or reported as a measured device result.

NSGA-II population search. The three-objective optimization uses a population P_t of size $N_{\text{pop}} = 80$, initialized uniformly over phase genes $[0, 1)^2$ and parameter intervals $[3.57, 4.0) \times [0, 0.5)$. Each generation performs nondominated sorting, crowding-distance assignment, binary tournament selection by rank then crowding distance, uniform per-gene crossover (probability 0.5), Gaussian mutation ($\sigma = 0.01$) clipped to the search bounds, and elitist $(\mu + \lambda)$ survivor selection back to 80 individuals. The run uses $G = 20$ generations.

FPA perturbation ablation. For the hybrid variant, every three generations up to ten members of the first nondominated front, ordered by crowding distance, are perturbed with the Flower Pollination Algorithm [8]. With probability $p = 0.8$, a global Lévy-flight perturbation moves the chromosome relative to a randomly chosen other population member; otherwise a Gaussian local perturbation is applied. The step uses Mantegna's construction with $\beta = 1.5$ and scale 0.01. The comparison variant runs the same algorithm and settings with these perturbations disabled.

Simulator and reproducibility. All optimization results are simulated, not measured on physical hardware. The source code, seed-level JSON outputs, and software versions are provided with the reproducibility materials. No device runtime, energy, or side-channel measurements are claimed. All runs use population size 80, 20 generations, and parameter bounds phase genes $p_x, p_y \in [0, 1)$ mapped as $x_0 = \sin^2(\pi p_x)$ and $y_0 = \sin^2(\pi p_y)$, $\mu \in [3.57, 4.0)$, and $\gamma \in [0, 0.5)$; paired optimizer runs use seeds 42, 100, 1234, 2025, and 7777.

Post-hoc S-box metrics. We report minimum and mean coordinate-function nonlinearity, strict avalanche criterion (SAC), BIC-SAC deviation, differential uniformity (DU), and linear approximation probability (LAP). Nonlinearity is computed from the Walsh–Hadamard spectra of the coordinate Boolean functions. SAC is the mean of the 8×8 avalanche-dependence matrix. BIC-SAC deviation is the mean absolute deviation of pairwise avalanche-bit XOR probabilities from 0.5. DU is the maximum nontrivial entry in the 256×256 difference-distribution table. LAP is the maximum probability bias $|\Pr[a \cdot x = b \cdot S(x)] - 1/2|$ over nonzero masks a, b. For a bijective 8-bit S-box, output Shannon entropy is exactly 8 bits by construction and is therefore not treated as an informative result. NPCR and UACI are omitted because this study does not define or test a complete encryption scheme; these are not properties of an isolated S-box.

Comparisons. The previous comparison combined 4-bit and 8-bit S-boxes and included baselines whose implementation and provenance were not sufficiently documented for a like-for-like comparison. We therefore do not make quantitative baseline claims in this revision. The reported optimizer ablation compares NSGA-II with the same implementation augmented by FPA perturbations, using paired seeds and identical population and generation budgets.

Software environment. The verified reruns used Python 3.12.3, NumPy 1.26.4, Matplotlib 3.10.8, and SciPy 1.14.1. Each optimizer variant used population size 80, 20 generations, Gaussian mutation standard deviation 0.01, and seeds 42, 100, 1234, 2025, and 7777. FPA runs perturbed up to ten high-crowding members every three generations; paired NSGA-II-only runs used the same seeds and settings with FPA disabled. The source and raw JSON outputs are included in the reproducibility deposit materials.

3 Results

The five-seed results in this section were regenerated using the configuration and source code described above; individual seed outputs are included in the reproducibility materials.

Lyapunov-exponent scale check. For the selected solutions across both variants, the corrected largest-exponent estimates were between 0.667 and 0.694. At $\gamma = 0$ and $\mu = 4$, the map reduces to independent logistic coordinates and the exponent is close to $\log 2$, as expected. The previous values near 16.8 resulted from an incorrect tangent renormalization and are withdrawn.

Paired optimizer comparison. We ran NSGA-II and NSGA-II with periodic FPA perturbations under the same five seeds (42, 100, 1234, 2025, 7777), population size 80, and 20 generations. Table 1 reports selected solutions as mean (SD). FPA-minus-NSGA-II-only differences were -0.00295 for LE (95% paired bootstrap CI [-0.00854, 0.00332]), -0.8 for $\mathrm{NL}_{\min}$ ([-1.6, 0.0]), 0.00239 for SAC ([-0.00107, 0.00605]), -0.00067 for BIC-SAC deviation ([-0.00199, 0.00066]), 0 for DU ([0,0]), and 0.00625 for LAP ([-0.00469, 0.01719]). Exact paired sign-flip tests with Holm correction gave adjusted $p = 1.0$ for all endpoints. With five seed pairs, these estimates are imprecise and do not establish superiority or equivalence. Mean final nondominated-front sizes were 6.0 with FPA and 4.4 without FPA; this count is descriptive and does not establish superior front quality.

The selected-solution averages are not a substitute for comparison of full Pareto-front quality. Future work should use a prespecified hypervolume or coverage measure and a larger independent seed set. We make no claim that FPA improves the search.

S-box properties. Across five selected solutions per variant, mean $\mathrm{NL}_{\min}$ was 101.6 (SD 0.89) for FPA and 102.4 (SD 0.89) for NSGA-II-only; mean DU was 10.0 (SD 0) for both. These 8-bit boxes have lower $\mathrm{NL}_{\min}$ and higher differential and linear biases than the standard AES S-box under the same metric implementations. These properties describe S-boxes only; they do not establish the security of a complete cipher. No NPCR/UACI or hardware performance results are reported.

4 Discussion

This study evaluates a simulated S-box construction and a paired search ablation; it does not evaluate a complete encryption system, side-channel resistance, or target-device performance. The five-seed results provide no statistically supported evidence of an FPA benefit. The corrected Lyapunov estimates are on the expected scale and supersede the previous values caused by tangent-vector renormalization error. Selected solutions span $\gamma = 0.00045$ to 0.49159, including near-zero coupling; a one-dimensional comparison is needed to test whether the second state variable contributes. The descriptive front sizes do not establish improved front quality; future work should use a prespecified hypervolume or coverage measure and more seeds.

Using the same metric implementations, the standard 8-bit AES S-box has $\mathrm{NL}_{\min} = 112$, DU= 4, and LAP bias = 0.0625. The paired optimizer means were $\mathrm{NL}_{\min} = 101.6$ (FPA) and 102.4 (NSGA-II-only), DU= 10 for both, and LAP bias = 0.1422 and 0.1359, respectively. These component metrics indicate that the proposed permutations are weaker than AES on the listed criteria and must not be presented as replacement cipher components. NPCR and UACI were removed because no complete cipher, key schedule, or diffusion protocol was specified. No Cortex-M0+ runtime or energy measurements were found, so device-performance claims are withdrawn. The corrected code and data are prepared locally for public deposit; a repository DOI must be added before resubmission.

5 Conclusions

We corrected and evaluated a Fisher–Yates S-box generator driven by a modulo-one coupled Logistic map and compared NSGA-II with and without periodic FPA perturbations. The fitness optimizes three independent objectives: largest Lyapunov exponent, minimum coordinate nonlinearity, and differential uniformity. In five paired seeds, selected mean $\mathrm{NL}_{\min}$ was 101.6 (SD 0.89) with FPA and 102.4 (SD 0.89) without FPA; mean DU was 10.0 for both. No paired differences were statistically supported after Holm correction. The results provide no evidence that FPA improves the search. Both variants underperform the standard AES S-box on the reported 8-bit NLmin, DU, and LAP metrics. The work is simulation-only and makes no claim about complete-cipher security, device runtime, energy, or side-channel resistance. Public deposition and a DOI, a larger-seed study, a 1D-versus-2D comparison, and validated like-for-like recent baselines remain necessary before resubmission.

Data availability

The corrected simulator, analysis scripts, configuration, paired seed outputs, and figures reported here are publicly available in the GitHub repository rawnjith/adaptive-multi-objective-chaotic-sbox (current `main` branch). An updated Zenodo archive and DOI for this exact release are pending. No third-party dataset is used.

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Author contributions

R. S. designed the chaotic-map-based S-box construction, ran the simulator, performed the cryptographic metric evaluations and the baseline comparison, supervised the multi-objective optimisation framework, and drafted the manuscript. R. R. A., A. P. V. and K. A. J. contributed to manuscript revision. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Figure legends

Figure 1. Three-objective nondominated set for seed 42, projected onto largest Lyapunov exponent and minimum coordinate nonlinearity; color encodes differential uniformity. The red marker shows the normalized-distance-to-ideal compromise selection.

Figure 2. Selected S-box reshaped as a 16×16 table. Every output value $0, \ldots, 255$ appears exactly once (bijection). The heatmap color encodes the S-box output value.

Figure 3. SAC matrix (left) and per-output-bit mean avalanche probability (right) of the selected seed-42 S-box. The dashed line at 0.5 marks the ideal avalanche balance.

Figure 4. Difference distribution table (DDT) of the seed-42 selected S-box on a log(1+count) scale. Its maximum nontrivial entry is 10, hence DU = 10.

Figure 5. Example orbit of the modulo-one coupled Logistic map at $(\mu, \gamma) = (3.91, 0.21)$. This visualization is illustrative and does not quantify the selected solutions' invariant distributions.

Figure 6. Best-so-far largest Lyapunov exponent, minimum coordinate nonlinearity, and differential uniformity across 20 generations for the FPA run with seed 42. Curves are descriptive and do not establish an FPA effect.

Figure 7. Paired selected $\mathrm{NL}_{\min}$ values for NSGA-II and NSGA-II with FPA perturbations across five seeds. Lines connect seed-matched runs; they move in both directions and do not show a consistent FPA advantage.

Table 1. Five-seed paired optimizer comparison showing mean (SD), paired mean difference, percentile bootstrap 95% CI, and Holm-adjusted exact paired sign-flip test results.

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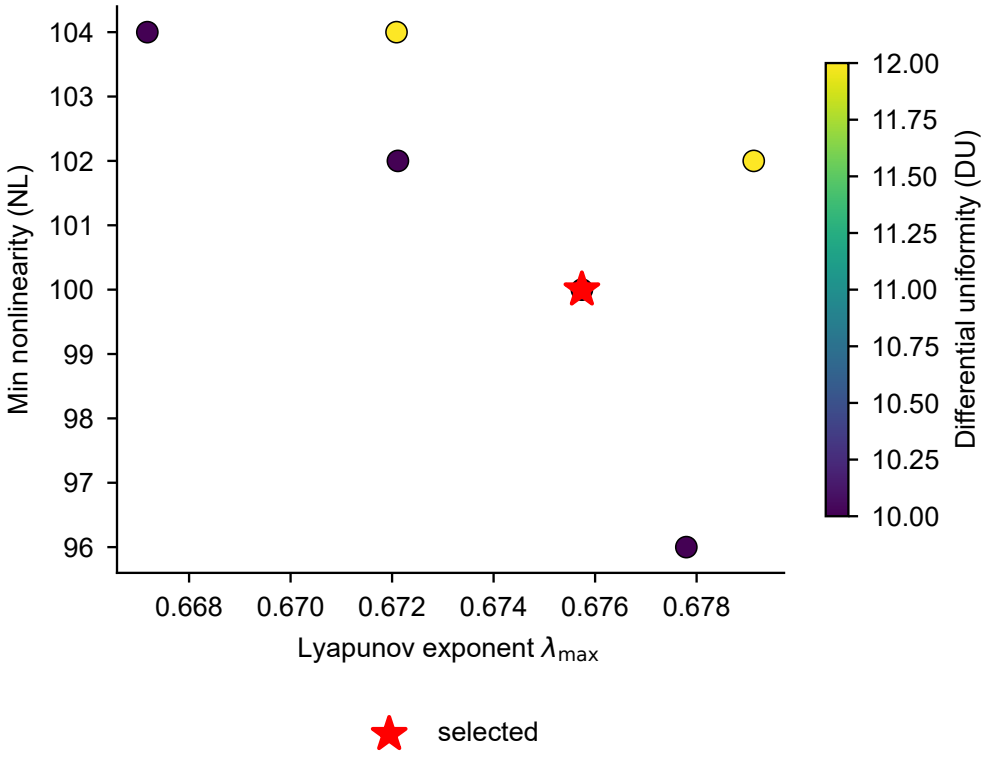


Figure 1: Three-objective nondominated set for seed 42, projected onto largest Lyapunov exponent and minimum coordinate nonlinearity; color encodes differential uniformity. The red marker denotes the normalized-distance-to-ideal compromise selection.

Table 1: Paired five-seed optimizer comparison. Entries are mean (SD); Δ is FPA minus NSGA-II-only, with percentile bootstrap 95% CI from 100,000 resamples of the five paired differences. Exact two-sided sign-flip tests were Holm-adjusted across six metrics; adjusted $p = 0.375$ for LE and $p = 1.00$ for each other metric.

Metric	FPA mean (SD)	NSGA-II mean (SD)	Δ [95% CI]
χ_{LE}	0.6725 (0.0029)	0.6754 (0.0078)	−0.00295 [−0.00854, 0.00332]
$\mathrm{NL}_{\mathrm{min}}$	101.6 (0.89)	102.4 (0.89)	−0.80 [−1.6, 0.0]
SAC	0.5006 (0.0034)	0.4982 (0.0040)	0.00239 [−0.00107, 0.00605]
BIC-SAC dev.	0.0342 (0.0008)	0.0348 (0.0016)	−0.00067 [−0.00199, 0.00066]
DU	10.0 (0.00)	10.0 (0.00)	0.00 [0.0, 0.0]
LAP	0.1422 (0.0065)	0.1359 (0.0089)	0.00625 [−0.00469, 0.01719]

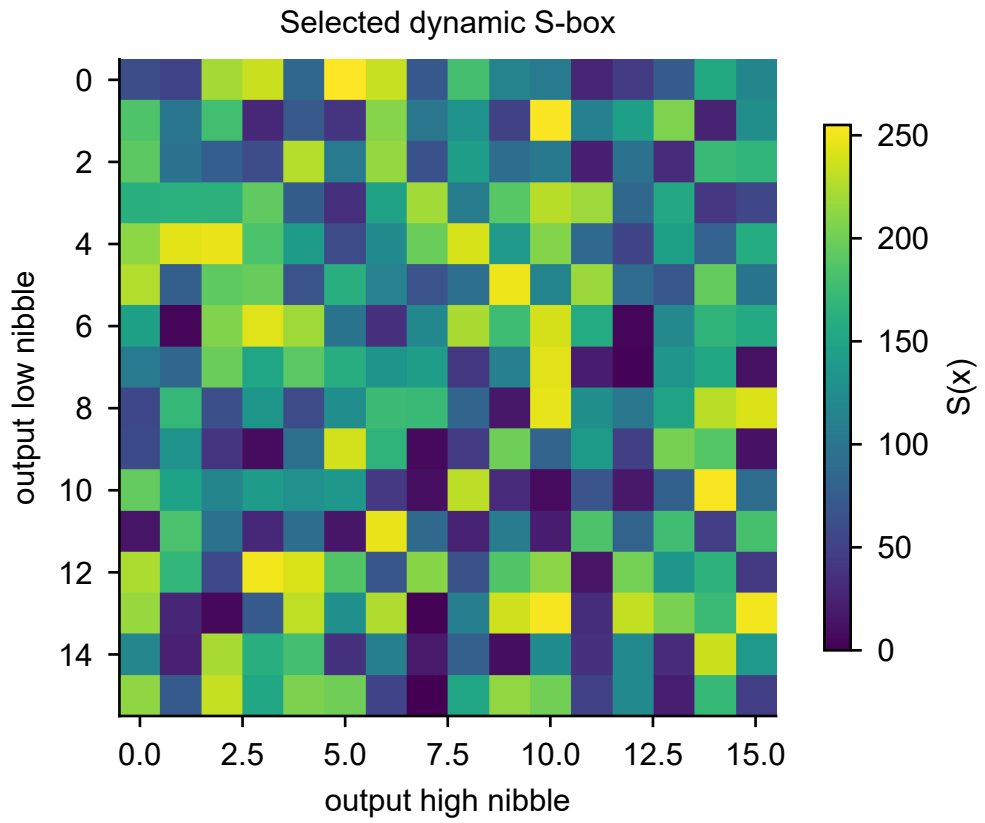


Figure 2: Selected S-box reshaped as a 16×16 table. Every output value $0, \ldots, 255$ appears exactly once (bijection).

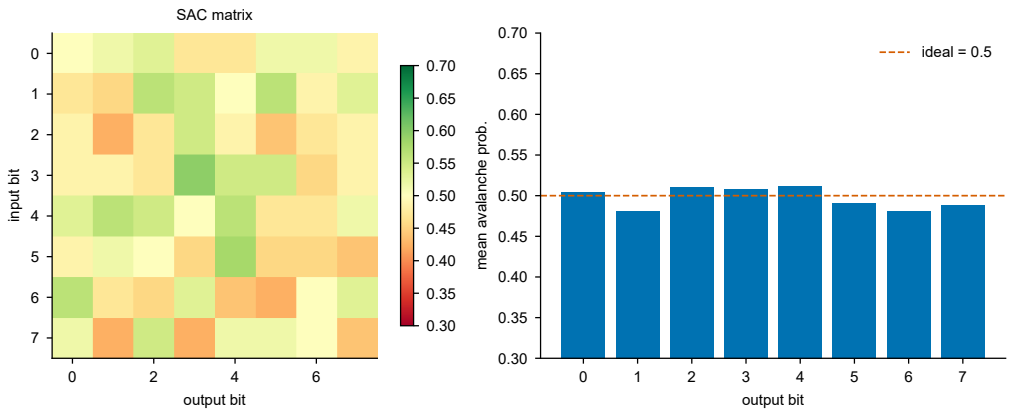


Figure 3: SAC matrix (left) and per-output-bit mean avalanche probability (right) of the seed-42 selected S-box. The dashed line at 0.5 marks the ideal avalanche balance. Overall SAC is 0.4966.

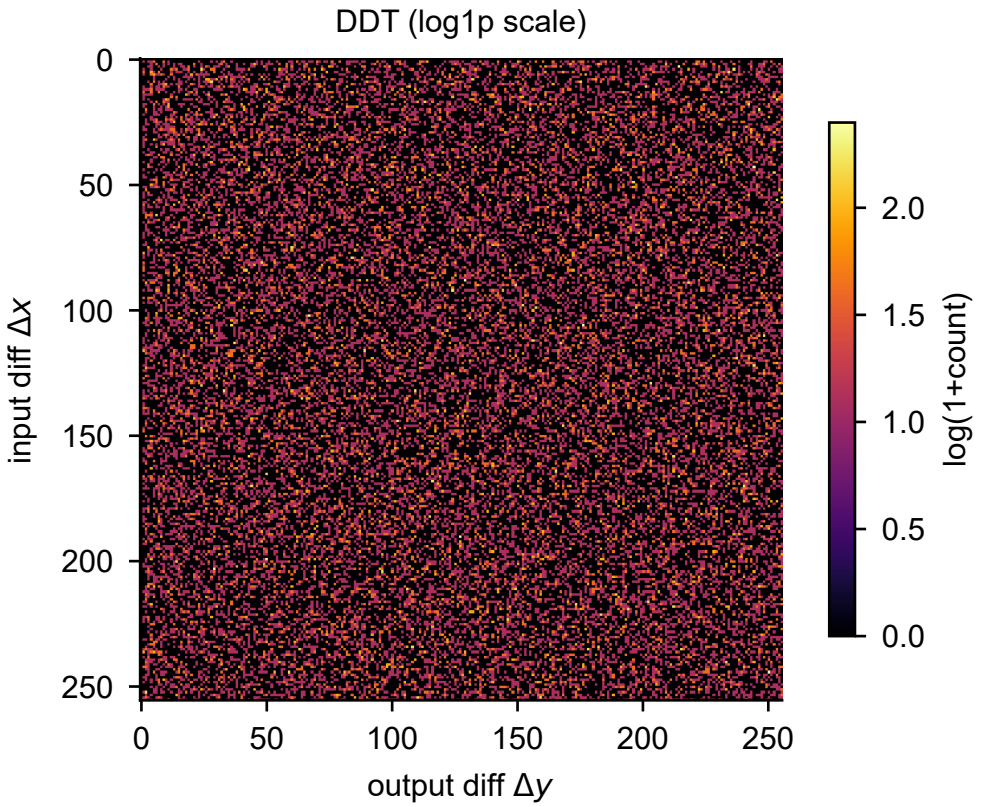


Figure 4: Difference distribution table (DDT) of the selected S-box on a $\log(1 + \text{count})$ scale. The peak DDT entry is 10, defining DU $= 10$.

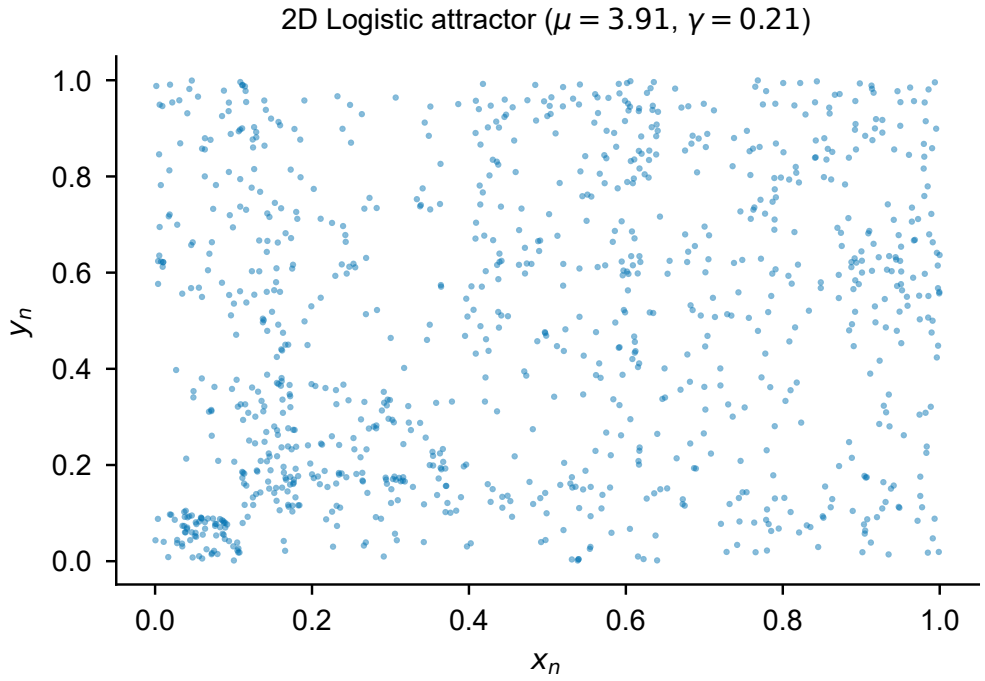


Figure 5: Illustrative orbit of the modulo-one coupled Logistic map at $(\mu, \gamma) = (3.91, 0.21)$; this is not a trajectory from a selected optimizer solution.

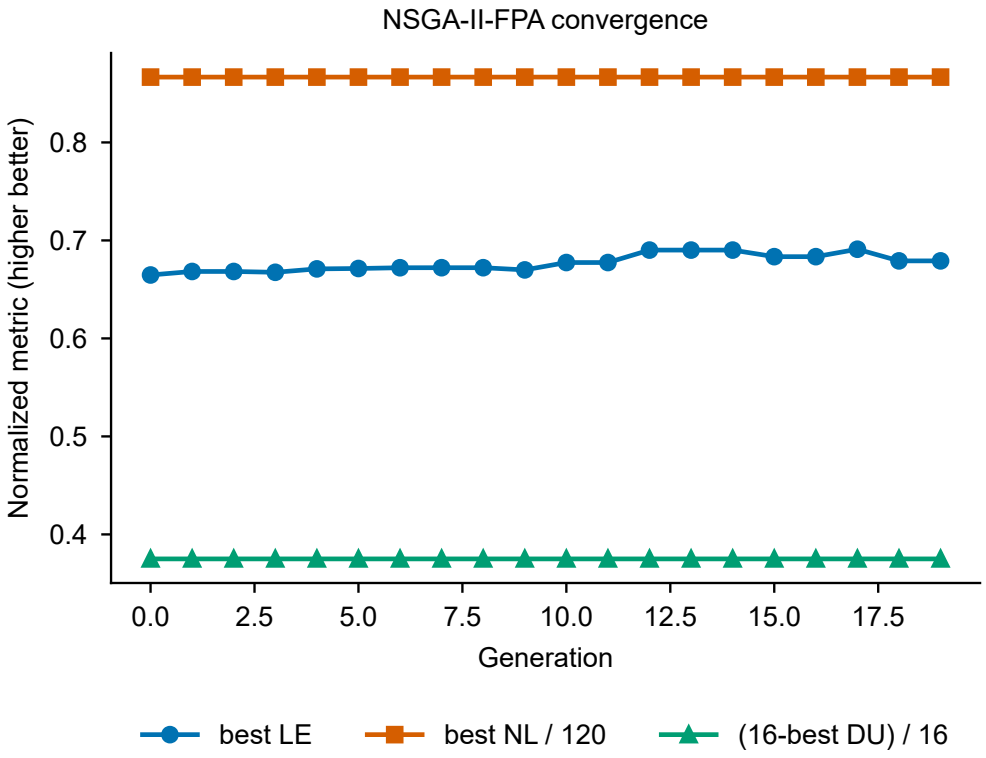


Figure 6: Best-so-far largest Lyapunov exponent, minimum coordinate nonlinearity, and differential uniformity over 20 generations for the FPA run with seed 42. Values are plotted on separate normalized scales; the curves are descriptive and do not establish an FPA effect.

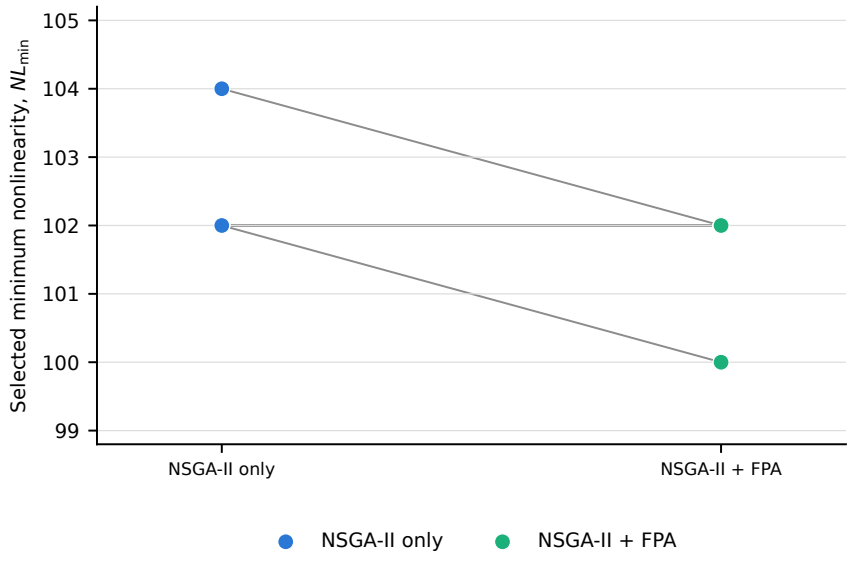


Figure 7: Paired selected NL_{min} values for NSGA-II and NSGA-II with FPA perturbations across five seeds. Lines connect runs sharing the same initialization seed. The overlap and mixed directions show that this small seed set does not establish an FPA advantage.